

focuses on the physical components of networking, specifically **interfaces and cables**, as part of the CCNA 200-301 curriculum. It covers the standards for copper and fiber optic cabling, how data is measured, and how devices communicate over these physical connections.

Network Protocols and Data Measurement

- **Protocols and Standards:** Networking requires agreed-upon systems (protocols) like **Ethernet** to ensure different vendors' equipment can communicate physically (connectors) and logically (IP).
- **Bits and Bytes:** Computer communication uses binary (0s and 1s).
 - **Speed** is measured in **bits per second** (e.g., Kbps, Mbps, Gbps), where a bit is a single 0 or 1.
 - **Data storage** is measured in **bytes**, where **1 byte equals 8 bits**.
 - A gigabyte is eight times larger than a gigabit.

Copper Ethernet Standards (UTP)

Most wired networks use **Unshielded Twisted-Pair (UTP)** cables with **RJ45 connectors**. These cables contain four pairs of wires twisted together to protect against electromagnetic interference (EMI).

Standard	Common Name	Speed	Max Length
10BASE-T	Ethernet	10 Mbps	100 meters
100BASE-T	Fast Ethernet	100 Mbps	100 meters
1000BASE-T	Gigabit Ethernet	1 Gbps	100 meters
10GBASE-T	10 Gig Ethernet	10 Gbps	100 meters

Source:

Cabling Types and Pinouts

The way wires are arranged in an RJ45 connector determines the cable type.

- **Pin Usage:** 10/100BASE-T uses two pairs (4 wires), while 1000BASE-T and higher use all four pairs (8 wires).
- **Device Pinouts:**
 - **PCs and Routers** transmit data on pins 1 & 2 and receive on pins 3 & 6.
 - **Switches** receive on pins 1 & 2 and transmit on pins 3 & 6.
- **Cable Types:**
 - **Straight-through Cable:** Connects different device types (e.g., PC to Switch) by mapping pin 1 to pin 1, etc.
 - **Crossover Cable:** Connects similar device types (e.g., Switch to Switch or Router to Router) by crossing the transmit and receive pairs.

- **Auto MDI-X:** This modern feature allows a port to automatically detect and adjust its pin usage, allowing devices to communicate normally regardless of whether a straight-through or crossover cable is used.

Fiber Optic Cabling

Fiber optic cables transmit data as **light pulses** over glass fibers rather than electrical signals over copper. They require **SFP (Small Form-factor Pluggable) transceivers** to connect to network devices.

- **Multimode Fiber (MMF):**
 - Has a wider core allowing multiple angles (modes) of light.
 - Uses cheaper LED-based transmitters.
 - Supports distances longer than UTP but shorter than single-mode.
- **Single-mode Fiber (SMF):**
 - Has a narrower core allowing only a single angle of light.
 - Uses more expensive laser-based transmitters.
 - Supports significantly longer distances (up to 30km or more).

Comparison Summary

Feature	UTP (Copper)	Fiber Optic
Cost	Cheaper	More expensive
Distance	Short (Max 100m)	Long (up to 30km+)
EMI	Vulnerable to interference	Immune to EMI
Security	Slight risk of signal leakage	No signal leakage; more secure
Port Type	RJ45	SFP

In practical scenarios, **UTP** is the standard for connecting end hosts to switches on the same floor. **Multimode fiber** is often used to connect switches between buildings at moderate distances (e.g., 150m), while **single-mode fiber** is necessary for long-distance connections (e.g., several kilometers).